

MiniProject 4: Learning with Sequence Data (LSTM vs Transformer)

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Abstract

This project compares two sequence modeling approaches for AG News classification: a bidirectional LSTM trained from scratch and a fine-tuned DistilBERT model. The LSTM uses word-level tokenization, manual vocabulary construction, padding, and masked mean pooling. DistilBERT uses a pretrained subword tokenizer and contextual representations. The LSTM achieves 92.29% test accuracy, while DistilBERT reaches 94.16%. Additional experiments compare recurrent variants, frozen vs full fine-tuning, LoRA, and the effect of sequence length on Transformer efficiency.

Introduction

We use the AG News dataset (120,000 train / 7,600 test examples, four classes: World, Sports, Business, Sci/Tech). The two pipelines differ mainly in preprocessing: the LSTM requires manual word-level tokenization and vocabulary construction, while the Transformer uses its pretrained subword tokenizer. This affects how rare words, padding, and context are handled. LSTMs process tokens sequentially; Transformers use self-attention across the full sequence. We use a 90/10 train/validation split (108,000 / 12,000) from the official training set, keeping the test set untouched until final evaluation.

Methods

LSTM pipeline

We use lowercase regex-based tokenization and build a vocabulary from training data only. Words with frequency ≤ 2 are removed; PAD and UNK tokens are added, yielding 34,306 tokens. Sequences are truncated to 128 tokens and padded per batch. We use `pack_padded_sequence` so padding is skipped during the LSTM forward pass. The model has an Embedding layer (dim 128), a bidirectional LSTM (hidden 256, 1 layer, dropout 0.3), masked mean pooling over real tokens, and a linear classifier (4 classes). Training: Adam (lr = 10^{-3}), batch size 64, 6 epochs, cross-entropy loss, gradient clipping at 1.0. Total parameters: 5,183,748.

Transformer pipeline

We fine-tune `distilbert-base-uncased` (66,956,548 params). The pretrained subword tokenizer adds [CLS] / [SEP] automatically. Sequences are truncated to 128 tokens and padded dynamically with attention masks (1 = real, 0 = PAD). The [CLS] hidden state is used for classification. Training: AdamW (lr = 2×10^{-5} , weight decay 0.05), batch size 16, 3 epochs, gradient clipping at 1.0.

Tokenization comparison

On the sentence *“Tyrannosaurus rex achieved its massive size due to an enormous growth spurt.”*:

Word-level: ['tyrannosaurus', 'rex', 'achieved', ..., 'spurt'] (12 tokens). “Tyrannosaurus” is one token; if rare, it maps to UNK and all meaning is lost.

DistilBERT subword: ['ty', '##ran', '##nos', '##aurus', 'rex', 'achieved', ..., 'spur', '##t', '.'] (17 tokens). Rare words are split into recognizable subword pieces; no token is ever UNK.

Beyond the minimum

We include: (1) recurrent ablations (pooling, directionality, GRU vs LSTM); (2) frozen-encoder vs full fine-tuning; (3) LoRA; (4) sequence-length trade-off analysis.

Table 1: Hyperparameter summary

Parameter	LSTM	Transformer
Tokenizer	Word-level regex	DistilBERT subword
Vocab size	34,306	30,522
Max seq. length / Hidden	128 / 256	128 / 768
Architecture	BiLSTM + mean pool	DistilBERT + [CLS]
Optimizer / LR	Adam / 10^{-3}	AdamW / 2×10^{-5}
Batch / Epochs / Dropout	64 / 6 / 0.3	16 / 3 / 0.1
Grad. clip / Params	1.0 / 5.18M	1.0 / 66.96M

Results and Analysis

Main performance

The LSTM reaches 92.30% validation and 92.29% test accuracy. DistilBERT achieves 94.47% peak validation and 94.16% test accuracy. Frozen-encoder and LoRA used reduced data (20k examples, 1 epoch) due to GPU constraints, so their lower scores partly reflect less training.

Table 2: Direct comparison of main models

Model	Val Acc.	Test Acc.	Params	Trainable	Total Time
LSTM baseline	92.30%	92.29%	5.18M	100%	3 min
DistilBERT full FT	94.47%	94.16%	66.96M	100%	51 min
DistilBERT frozen*	—	89.25%	66.96M	0.89%	5 min
DistilBERT LoRA*	—	89.45%	67.85M	1.31%	2 min

*Trained on 20k examples, 1 epoch. All times on Colab T4 GPU.

Training behavior

The LSTM trains stably over 6 epochs: training loss drops from 0.494 to 0.142, validation accuracy rises from 89.97% to 92.30%, with mild overfitting visible as validation loss increases slightly in later epochs. DistilBERT converges much faster: training loss drops from 0.163 to 0.083 over 3 epochs, validation accuracy peaks at 94.47% (epoch 2) then drops to 94.31%, with validation loss rising sharply in epoch 3.

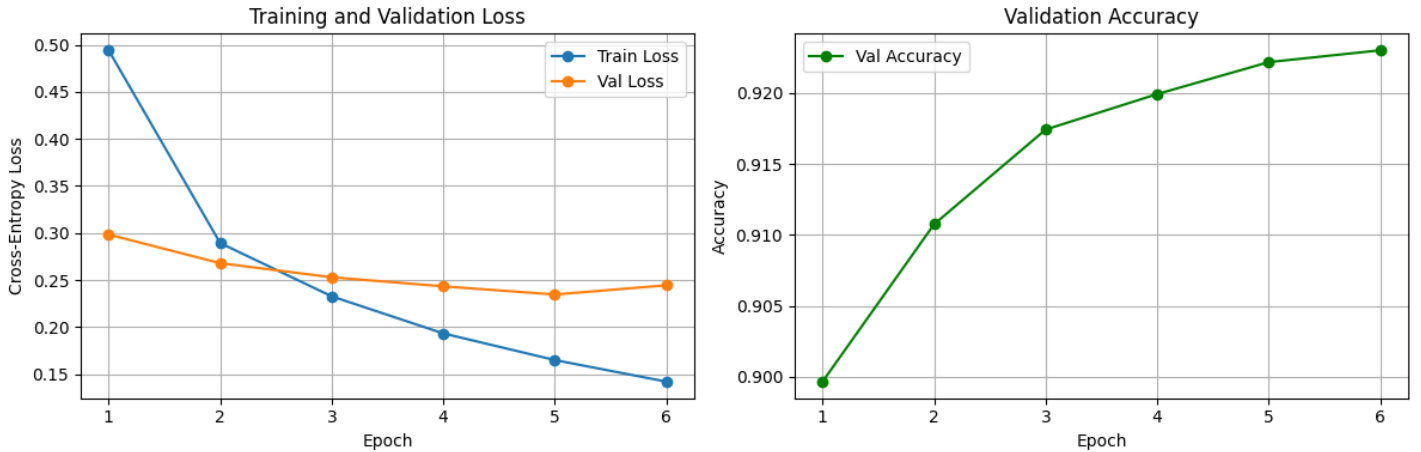


Figure 1: LSTM learning curves: loss (left) and validation accuracy (right).

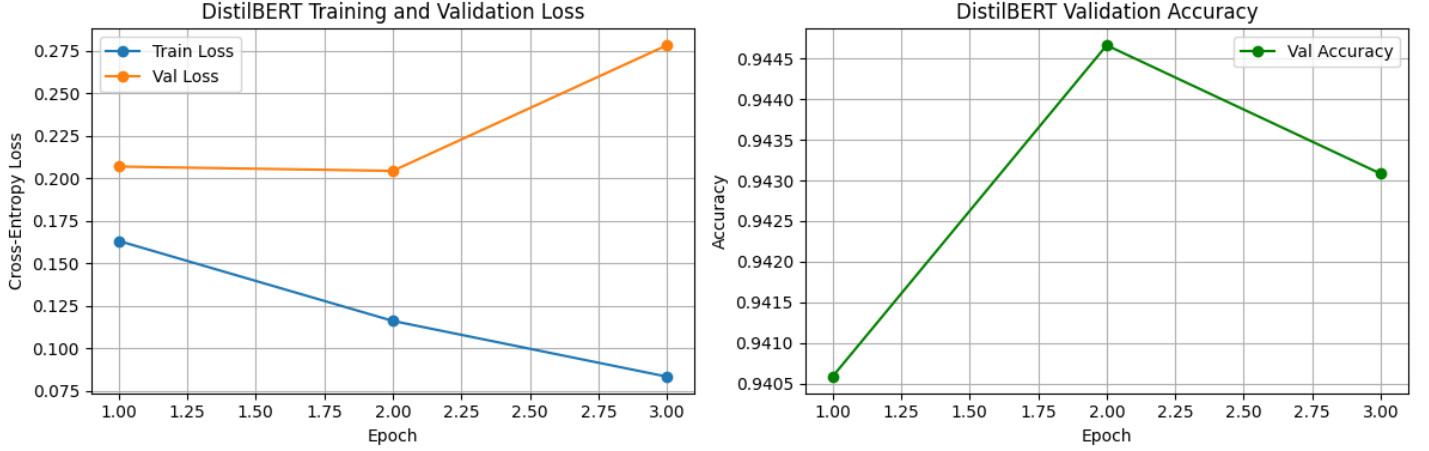


Figure 2: DistilBERT learning curves. Overfitting begins after epoch 2.

Recurrent variant ablation

We compare RNN variants with fixed preprocessing and optimization, re-seeding before each run for fair comparison.

Table 3: RNN ablation (all 6 epochs, full training set)

Configuration	Test Acc.
BiLSTM + mean pooling (baseline)	92.21%
BiLSTM + last hidden state	92.04%
Unidirectional LSTM + mean pooling	91.91%
BiGRU + mean pooling	92.09%

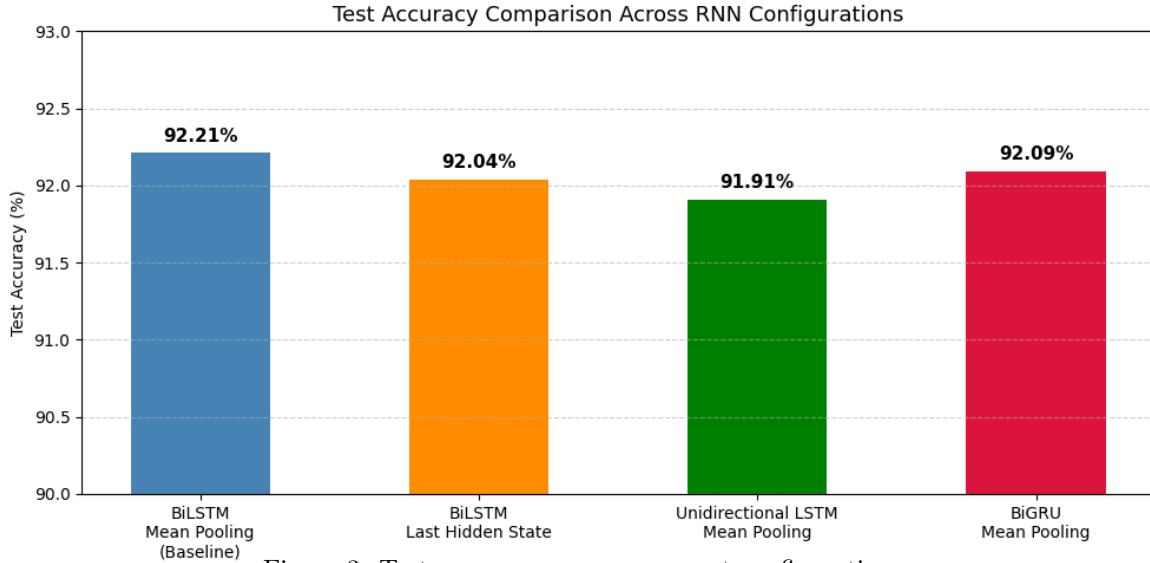


Figure 3: Test accuracy across recurrent configurations.

Parameter-efficient fine-tuning

Full fine-tuning outperforms frozen-encoder and LoRA. The frozen/LoRA experiments used reduced data (20k examples, 1 epoch), so the gap partly reflects less training, but the results still suggest AG News benefits from updating the encoder.

Table 4: Fine-tuning strategy comparison

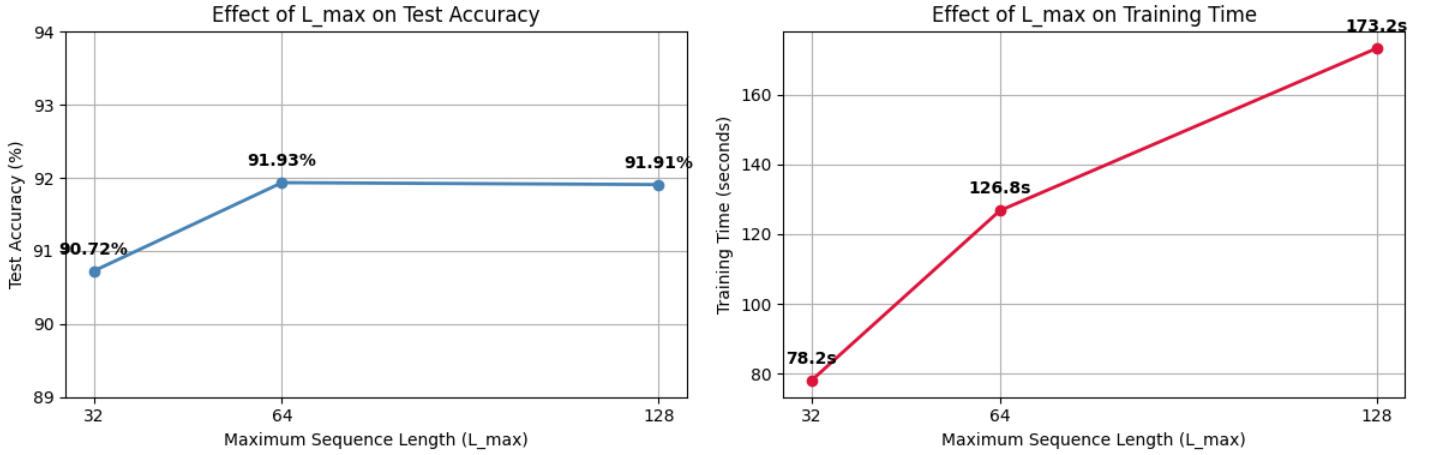
Setup	Trainable	%	Train Data	Test Acc.
Full fine-tuning	66.96M	100%	108k, 3 ep.	94.16%
Frozen encoder	593k	0.89%	108k, 1 ep.	89.25%
LoRA ($r=16$)	889k	1.31%	20k, 1 ep.	89.45%

Effect of sequence length

We trained DistilBERT with $L_{\max} \in \{32, 64, 128\}$ on 20k examples for 1 epoch. Accuracy improves from 32 to 64 tokens but plateaus at 128, while training time increases by 37%.

Table 5: Sequence length trade-off (20k subset, 1 epoch)

$L_{\max}$	Test Acc.	Time (s)
32	90.72%	78.2
64	91.93%	126.8
128	91.91%	173.2

Figure 4: $L_{\max}$ vs test accuracy (left) and training time (right). Accuracy plateaus at 64.

Qualitative error analysis

Table 6: Sample LSTM misclassifications (8 of 587 total errors)

Text (truncated)	True	Pred.
Prediction Unit Helps Forecast Wildfires (AP)...	Sci/Tech	Sports
Card fraud unit nets 36,000 cards...	Sci/Tech	Business
Rivals Try to Turn Tables on Charles Schwab...	Sci/Tech	Business
Promoting a Shared Vision. As Michael Kaleko kept running into people...	Sci/Tech	Business
Venezuela Prepares for Chavez Recall Vote...	World	Business
Haas and Cink Selected for Ryder Cup Team...	Sports	World
Intel to delay product aimed for high-def TVs...	Business	Sci/Tech
Oil prices bubble to record high...	World	Business

Table 7: Sample DistilBERT misclassifications (8 of 444 total errors)

Text (truncated)	True	Pred.
Card fraud unit nets 36,000 cards...	Sci/Tech	World
Some People Not Eligible to Get in on Google IPO...	Sci/Tech	Business
Intuit Posts Wider Loss After Charge (Reuters)...	Sci/Tech	Business
Intel to delay product aimed for high-def TVs...	Business	Sci/Tech
Sneaky Credit Card Tactics...	Business	Sci/Tech
Yahoo! Ups Ante for Small Businesses...	Business	Sci/Tech
Future Doctors, Crossing Borders...	World	Sci/Tech
Afghan women make brief Olympic debut...	Sports	World

The dominant error pattern for both models is Sci/Tech $\leftrightarrow$ Business confusion: 5 of 8 LSTM errors and 6 of 8 DistilBERT errors shown above fall on this boundary. These articles discuss technology companies, IPOs, or products that genuinely span both categories. The LSTM is additionally prone to confusing World $\leftrightarrow$ Business when articles mention oil prices or political economy, likely because financial vocabulary dominates and word-level tokenization loses rare geopolitical terms to UNK. The Transformer makes fewer total errors (444 vs 587) and its remaining mistakes tend to involve genuinely ambiguous content, such as the Olympics article labeled Sports but discussing Afghan women (a World angle) or medical education labeled World but touching on science. Notably, both models misclassify the “Card fraud” article, but to different wrong classes (Business vs World), illustrating how each model’s representation leads to distinct decision boundaries.

Discussion and Conclusion

The Transformer achieves the best accuracy (94.16%) via subword tokenization, pretrained representations, and self-attention, but at 13x more parameters and 7x longer training per epoch. The LSTM reaches 92.29% with only 5.18M parameters. Among recurrent variants, bidirectional processing and mean pooling were consistently best, though all configurations fell within 0.3pp. Full fine-tuning outperformed frozen-encoder and LoRA, confirming that adapting pretrained representations matters. The sequence-length experiment showed no benefit beyond 64 tokens. A natural next step would be initializing the LSTM embedding layer with GloVe or FastText vectors to reduce the accuracy gap while preserving simplicity.

Statement of Contributions

Shahmeer Sajid was responsible for the code (submitted as code.ipynb), while the report (submitted as writeup.pdf) was written by Danyal Valika and Uday Parmar. All team members reviewed both the code and the report before submission.

Statement on the Use of LLMs and Other Resources

Large language models were used for guidance on report structuring and LaTeX formatting to fit the 5 page limit. The LLM was not used to implement the core algorithms or report writing. No code or data was shared with other teams.